

EE2020/EE2026 (Part 1)

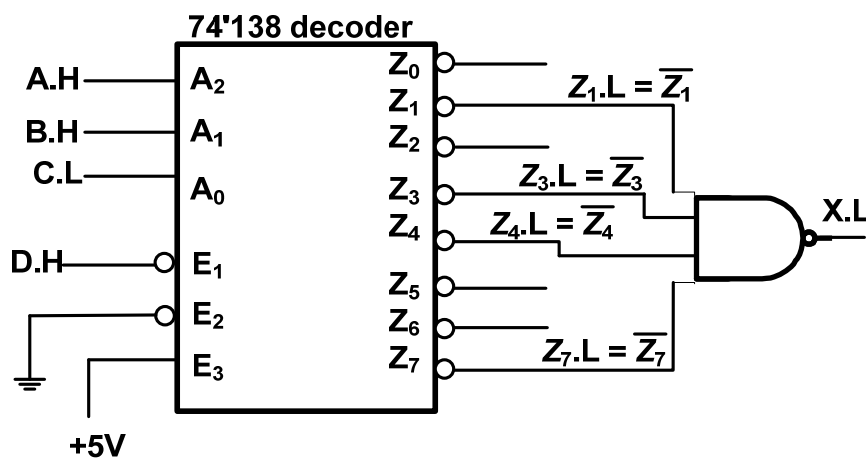
Tutorial 5 - Questions

Combinational MSI circuits, building blocks and Verilog

- Use a 74'283 adder chip and some logic gates to find the magnitude of a 4-bit number represented in the 2's complement notation.

Write the Verilog dataflow code to describe the same function.

- Find the Boolean expression for X in the circuit below.



- $S_3S_2S_1S_0$ is a 4-bit number in the 2's complement representation, and **S** is the corresponding decimal number.
 - Write down a table showing the value of **S** for all combinations of $S_3S_2S_1S_0$.
 - It is required to compare two 4-bit signed numbers, **A** and **B**, represented in the 2's complement notation. If a 4-bit *magnitude* comparator is available, design a circuit based on this comparator to yield the relationship between the two *signed* numbers (i.e. greater than, less than, equal). Show a neat, clearly labeled circuit diagram and give a brief explanation of your design (*Hint*: use gates?). Finally, write the Verilog dataflow description of this function.
- Z is a Boolean function of 3 input variables A, B and C. $Z = B$, if $A = C$; else $Z = 0$.
 - Design and draw a clearly labeled circuit diagram to implement Z using only one 8-bit multiplexer.
 - Can problem in part (a) be done with only one 4-bit MUX instead of the 8-bit MUX?
 Finally, write the Verilog dataflow description of the intended function.
- Table below shows the relationship between two 4-bit codes B and N. Design a circuit to convert from code B to code N. For example, if the input B codeword is

“0001”, the circuit should output N codeword “0111” by using some combination of decoders, encoders, multiplexers, de-multiplexers and comparators, without using any gates at all. Draw a neatly labeled circuit diagram and provide a brief explanation of circuit operation.

B₃	B₂	B₁	B₀	N₃	N₂	N₁	N₀
0	0	0	0	0	0	1	1
0	0	0	1	0	1	1	1
0	0	1	0	1	0	0	0
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	1	0	0	1
0	1	1	0	0	0	0	0
0	1	1	1	1	1	1	1
1	0	0	0	0	1	0	1
1	0	0	1	0	0	0	1
1	0	1	0	1	0	1	0
1	0	1	1	0	1	0	0
1	1	0	0	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	0	1	1	0	0
1	1	1	1	1	1	1	0